

Japanese Zelkova

Zelkova serrata

Species background: Native to Japan, Taiwan, and parts of eastern China, often planted as a substitute for American elm because of its similar vase-shaped crown. Its vase-shaped, upward-arching branch structure closely echoes the American elm's classic silhouette, which is exactly why cities began planting it widely after Dutch elm disease killed so many elms.

This packet contains one full lesson for each grade band: K-2, 3-5, 6-8, and 9-12. Each includes a learning objective, standards connection, materials, teacher background, a step-by-step procedure, discussion questions, an assessment, and an extension activity.

GRADE BAND K-2

Japanese Zelkova: Identify zelkova's cup-shaped branching and find the tiny lines on its bark.

Standards connection: NGSS K-LS1-1; National Core Arts Standards VA:Cr1 (K-2)

Materials: Coloring sheet, crayons.

Teacher background

Step back and look up at this tree -- do its branches make a wide cup or vase shape? Zelkova trunks also have tiny horizontal lines called lenticels, small pores the tree uses to breathe.

Procedure

- 1 Engage: Ask students to breathe in and out, then ask how they think a tree without a nose breathes.
- 2 Explore: Look up at the tree's branches together and trace the cup shape in the air.
- 3 Explain: Introduce lenticels as the tree's tiny breathing holes on the bark.
- 4 Art: Look up at the branches and draw their shape with one continuous line.
- 5 Evaluate: Student can point out the vase-shaped branches and name lenticels as breathing holes.

Discussion questions

What other living things have a special way of breathing that's different from ours?

Assessment

Correctly points to the vase shape and recalls the word 'lenticels' with help.

Extension

Find another tree's bark and look for its own small marks or bumps.

GRADE BAND 3-5

Japanese Zelkova: Explain what a lenticel does and describe zelkova's crown shape as a light-gathering strategy.

Standards connection: NGSS 4-LS1-1; National Core Arts Standards VA:Cr2 (3-5)

Materials: Coloring sheet, colored pencils.

Teacher background

Zelkova's vase-shaped crown is a light-competition strategy for trees planted close together along streets, letting each tree's leaves reach open sky above its neighbors' shade.

Procedure

- 1 Engage: Ask why two trees planted right next to each other might grow their branches in different directions.
- 2 Explore: Compare zelkova's branch shape to a nearby round-crowned tree.
- 3 Explain: Discuss lenticels as gas-exchange structures and the vase shape as a light-gathering strategy.
- 4 Art: Sketch just the outline of the tree's branches against the sky, no leaves -- vase-shaped branches make a great line drawing.
- 5 Evaluate: Student writes one sentence about why the vase shape might help this tree get more sunlight.

Discussion questions

Would this vase shape still be useful for a zelkova growing alone in an open field?

Assessment

Correct sentence connecting crown shape to sunlight access.

Extension

Sketch two different tree crown shapes seen on your walk and label which might catch more light.

Japanese Zelkova: Compare crown shapes as light-competition strategies among crowded street trees.

Standards connection: NGSS MS-LS1-1; National Core Arts Standards VA:Cr2/Re7 (6-8)

Materials: Coloring sheet, pencil, optional protractor.

Teacher background

Zelkova's vase-shaped crown reduces self-shading among crowded street trees. Its lenticels are gas-exchange structures in the bark, and the tree is naturally resistant to the Dutch elm disease that devastated American elm, which is exactly why cities began planting zelkova as a substitute.

Procedure

- 1 Engage: If two trees compete for the same patch of sunlight, what strategies might each one use to win?
- 2 Explore: Sketch the tree's vase-shaped silhouette, then compare it to a nearby tree's silhouette.
- 3 Explain: Label which shape captures more sunlight and why, and connect zelkova's disease resistance to its role as an elm replacement.
- 4 Art: Sketch the tree's vase-shaped silhouette, then compare it to a nearby tree's silhouette -- label which shape captures more sunlight and why.
- 5 Evaluate: Write a short paragraph comparing the two crown shapes' light-gathering strategies.

Discussion questions

Why might cities intentionally choose disease-resistant substitute species instead of replanting the original tree?

Assessment

Paragraph correctly compares crown shapes and explains the light-competition reasoning.

Extension

Research one other tree species used as a Dutch-elm-disease substitute besides zelkova.

Japanese Zelkova: Measure branching geometry and evaluate crown architecture as a light-competition strategy.

Standards connection: NGSS HS-LS2-6; National Core Arts Standards Proficient-level Creating/Responding (9-12)

Materials: Coloring sheet, tape measure, protractor.

Teacher background

Measure the tree's diameter at breast height (DBH) and estimate age using a species growth-factor formula, then compare its growth rate to a same-age tree with a different crown shape. Vase-shaped branching reduces self-shading among crowded street trees, a real architectural strategy for competing for light in dense plantings.

Procedure

- 1 Engage: How might you quantify a tree's 'strategy' for capturing light, rather than just describing it?
- 2 Explore: Measure DBH and estimate the tree's age, then measure the branching angle at two major forks.
- 3 Explain: Research crown architecture and light competition in urban forestry, and note whether the branching angle stays consistent across the tree.
- 4 Art: Produce a contour-line study of the branching structure, marking each major fork and labeling measured angles.
- 5 Evaluate: Submit the DBH/age estimate, angle measurements, and the labeled contour drawing.

Discussion questions

Does a consistent branching angle suggest a genetically programmed growth pattern, or a response to this specific tree's light environment -- or both?

Assessment

Rubric covering measurement accuracy, research depth, and drawing precision.

Extension

Compare zelkova's disease-resistance strategy to Chinese elm's -- both were adopted as American elm substitutes for similar reasons.

SOURCES & METHODOLOGY

Background research drawn from TreeWalk NYC's own species research. Lesson structure (objective, background, materials, procedure, assessment, extension) follows the format used by real, freely published environmental-education lessons, including the National Park Service's "Looking at Leaves" 5E lesson plan (Acadia Outdoor Classroom Collaborative), Penn State Extension's "Sorting and Classifying Leaves," and Project Learning Tree's leaf-activity guidance -- adapted and written fresh for this species, not copied from those sources. This is an educational pilot; standards references indicate topic alignment, not formal certification.